

Sara Mascia

PhD, Curriculum Vitae

Research Interests

- **General:** Astronomy outreach
- **Astronomical instrumentation, methods and techniques:** Telescopes (ultraviolet, optical, infrared), photometry, spectroscopy
- **Galaxies:** active, distances and redshifts, evolution, high-redshift
- **Cosmology:** observations, early Universe

Education

- 2024 – Present **ISTA Prize Fellow**, affiliated with the Matthee group, Institute of Science and Technology Austria (ISTA)
- 2021 – 2024 **PhD in Astronomy, Astrophysics and Space Science**, University of Rome "Tor Vergata" with the collaboration of the University of Rome "Sapienza" and the National Institute of Astrophysics (INAF) + **Amaldi Research Center (ARC) fellow**
- **PhD Thesis:** "Tracing the properties of Lyman continuum emitting galaxies over cosmic epochs" under supervision of **Prof. Laura Pentericci** and **Dr. Marco Castellano**
- 2019 – 2021 **M.Sc. cum Laude in Astrophysics and Space Physics**, LM-58, University of Milano Bicocca, Italy
- **M.Sc. Thesis:** "Star formation and quenching of low mass galaxies at $z \sim 1 - 2$ in the MUSE Ultra Deep Field (MUDF) survey", under supervision of **Prof. Michele Fumagalli** and **Dr. Matteo Fossati**
- 2016–2019 **B.Sc. cum Laude in Physics**, L-30, Università degli Studi di Cagliari, Italy
- **B.Sc. Thesis:** "Analisi temporale del pulsatore X al millisecondo XTE J0929-314", under supervision of **Prof. Alessandro Riggio**

International schools attended

- Oct. 2023 **4th Italian Astrostatistics School**, INAF - Osservatorio Astronomico di Brera, Milano
- Sept. 2023 **18th Heidelberg Summer School on Unraveling Galaxy Evolution with JWST**, Max Planck Research School for Astronomy and Cosmic Physics at the University of Heidelberg (IMPRS-HD)
- May 2022 **IAA-CSIC Severo Ochoa Advanced School on Galaxy Evolution**, Instituto de Astrofísica de Andalucía headquarters (IAA-CSIC), Granada
- Feb. 2022 **DAWN Winter School in Astrophysics**, University of Copenhagen, Faculty of Science

International collaborations

- 2021 – Present **VANDELS**, (PID 194.A-2003, PI L. Pentericci and R. McLure)
- 2022 – Present **GLASS-JWST-ERS**, (JWST-ERS-1324: PI T. Treu)
- 2022 – Present **FLAGS**, (First Light And Galaxy aSsembly, PI: L. Pentericci and R. Schneider)

- 2023 – Present **CEERS**, (*Cosmic Evolution Early Release Science survey, JWST-ERS-1345, PI: S. Finkelstein*)
- 2023 – Present **NGDEEP**, (*The Next Generation Deep Extragalactic Exploratory Public, JWST-GO-2079, PI: S. Finkelstein*)
- 2024 – Present **LyC22**, (*Deep spectroscopic insights on star-forming galaxies 2.2 Gyr after the Big Bang, JWST-GO-1869, PI: D. Schaerer*)

Observational programs

- Co-I of **To catch an AGN: Ionizing Emission and AGN Activity of COLA1 in the Epoch of Reionization**, GO 7722, PIs A. Torralba, J. Matthee (JWST)
- Co-I of **Searching for Population III stars around massive interacting galaxies in the reionization epoch**, GO 7014, PI A. Calabro' (JWST)
- Co-I of **A deep look into the physics of the interstellar medium 360 Myr after the Big Bang**, GO 7201, PI M. Castellano (JWST)
- Co-I of **The CANDELS-Area Prism Epoch of Reionization Survey (CAPERS)**, PI M. Dickinson (JWST)
- Co-I of **On the nature of UV emission line galaxies: a study of CIV emitters at the Cosmic Noon**, PI A. Saldana Lopez (XSHOOTER)
- Co-I of **Connecting Lyman Continuum emission to UV spectral properties in VANDELS galaxies at $z \sim 3.5$** , PI L. Pentericci (MODS-LBT)
- Co-I of **Spectroscopic confirmation of an unexpected population of bright galaxies at cosmic dawn**, GO 3073, PI M. Castellano (JWST)

Miscellaneous

- 2025 **A&A Award for Outstanding PhD Article**
- March 2024 **Visitor at Observatoire de Genève, Versoix, Switzerland**, Reduction and analysis of JWST/NIRSpec data. Collaboration with prof D. Schaerer.
- 2023 – Present **LBT Proposal Science Reviewer**
- Nov. – Dec. 2022 **Visitor at UCLA, Los Angeles, CA, USA**, Acquirement of skills in reduction and analysis of JWST/NIRSpec data. Collaboration with prof T. Treu and PhD. G. Roberts Borsani.

Conferences

- **Invited talk:** *A (not so) Complete Unknown: multiple predictors of Lyman Continuum escape in the early Universe*, Conference: Escape of Lyman radiation from galactic labyrinths, Kolymbari, Crete, Greece, 8-11 April 2025
- **Contributed talk:** *Understanding Lyman Continuum escape in high-redshift galaxies: insights from JWST observations*, Conference: Understanding the Epoch of Reionization II, Sesto, Val Pusteria, Italy, 13-17 January 2025
- **Contributed talk:** *The impact of mergers on the Lyman Continuum photons escape in Galaxies during the epoch of reionization*, Conference: Cosmic Dawn at High Latitudes Conference, Stockholm, Sweden, 24-28 June 2024
- **Invited talk:** *Unveiling the Cosmic Dawn: Exploring the High-Redshift Universe with JWST*, Physics Colloquium, Università di Cagliari, INFN, Sezione di Cagliari, Italy, 27 March 2024
- **Contributed talk:** *Escaping the labyrinth of reionization with JWST observations*, Conference: The growth of galaxies in the Early Universe - XIV, Sesto, Val Pusteria, Italy, 22-26 January 2024

- **Contributed talk:** *The indirect indicators' thread: escaping the labyrinth of reionization*, Conference: 12th Young Researcher Meeting, Monte Porzio Catone, Italy, 2-4 October 2023
- **Contributed talk:** *The indirect indicators' thread: escaping the labyrinth of reionization*, Conference: The James Webb Space Telescope turns one: the birth and growth of galaxies, Sesto, Val Pusteria, Italy, 10-14 July 2023
- **Contributed talk:** *The indirect indicators' thread: escaping the labyrinth of reionization*, Conference: Shedding new light on the first billion years of the Universe, Marseille, France, 3-7 July 2023
- **Contributed talk:** *The indirect indicators' thread: escaping the labyrinth of reionization*, Students' Day @ OAR, Roma, Italy, 9 May 2023
- **Contributed talk:** *Reionization's labyrinth: how to escape following the indirect indicators' thread*, Conference: Escape of Lyman radiation from galactic labyrinths, Kolymbari, Crete, Greece, 18-21 April 2023
- **Contributed talk:** *Closing in on the sources of cosmic reionization: first results from the GLASS-JWST program*, Conference: The growth of galaxies in the Early Universe - VIII, Sesto, Val Pusteria, Italy, 13-17 March 2023
- **Contributed talk:** *Insight into the reionization epoch from cosmic high-noon studies of UV emission galaxies*, European Astronomical Society Annual Meeting, Valencia, Spain, 27 June – 1 July 2022
- **Contributed talk:** *The VANDELS Survey: Properties of the CIV $\lambda 1550$ emitters at $z \sim 2.5 - 5$* , VANDELS final meeting, Bologna, Italy, 12-13 May 2022
- **Contributed talk:** *The VANDELS Survey: Properties of the CIV $\lambda 1550$ emitters at $z \sim 2.5 - 5$* , Students' Day @ OAR, Roma, Italy, 10 May 2022

Outreach

- 01/12/2023 **Author of the article in the INAF magazine UNIVERSI, entitled "Ecco le galassie che hanno reionizzato l'universo" (Italian)**
- 29-30/09/2023 **Partecipation at the "Notte dei Ricercatori 2023", Città dell'altra economia, Testaccio, Roma**
- 18/03/2023 **Cosmo 2050, radio interview, Giornale Radio**
- 13/03/2023 **Ecco le galassie che hanno reionizzato l'universo, press release, Media INAF (Italian), + 7 other articles on different italian newspapers based on it**

Teaching and tutoring

- 2023-2024 **Representative of PhD Candidates, University of Rome "Tor Vergata" with the collaboration of the University of Rome "Sapienza" and the National Institute of Astrophysics (INAF)**
- 2023 **PCTO INAF: external tutor, Liceo Scientifico Statale Vito Volterra, Ciampino (RM)**

Skills

- **Scientific Software:** MATLAB, LaTeX, DS9, QFitsView, Overleaf, PyMultinest, Pandora, Sippi
- **Operating Systems:** Linux, Mac OS X, Windows
- **Programming Languages:** Python/Python3, C, C++
- **Languages:** Italian (native), English (fluent)

1 Publication Statistics

1771 citations according to NASA ADS (14 April 2025)

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Highlights: Front page illustration of *Astronomy & Astrophysics*, Volume 672 (April 2023)

2 First Author/Major Contribution

1. **S. Mascia**, et al., Little impact of mergers and galaxy morphology on the production and escape of ionizing photons in the early Universe, submitted to *Astronomy and Astrophysics* 2025
2. **S. Mascia**, et al., The GLASS-JWST Early Release Science Program. IV. Data release of 263 spectra from 245 unique sources, *Astronomy and Astrophysics*, 690, A2 (2024)
3. **S. Mascia**, et al., New insight on the nature of cosmic reionizers from the CEERS survey, *Astronomy and Astrophysics*, 685, A3 (2024)
4. G. Prieto-Lyon, C. Mason, **S. Mascia**, et al., Early Results from GLASS-JWST. XXIII. The Transmission of Ly α from UV-faint z 3-6 Galaxies, *The Astrophysical Journal*, 956, 136 (2023)
5. **S. Mascia**, et al., Closing in on the sources of cosmic reionization: first results from the GLASS-JWST program, *Astronomy and Astrophysics*, 672, A155 (2023)
6. **S. Mascia**, et al., Insights into the reionization epoch from cosmic-noon C IV emitters in the VANDELS survey, *Astronomy and Astrophysics*, 674, A221 (2023)
7. K. Boyett, **S. Mascia**, et al., Early results from GLASS-JWST. VI. Extreme rest-optical equivalent widths detected in NIRISS Wide Field Slitless Spectroscopy, *The Astrophysical Journal*, 940, L52 (2022)

3 Other Publications

1. M. Castellano,..., **S. Mascia**, et al., Pushing JWST to the extremes: search and scrutiny of bright galaxy candidates at $z \sim 15 - 30$, submitted to *A&A*, (2025)
2. D. Dottorini,..., **S. Mascia**, et al., Evolution of the UV slope of galaxies at cosmic morning ($z > 4$): the properties of extremely blue galaxies, submitted to *A&A*, (2024)
3. B. Vulcani,..., **S. Mascia**, et al., Not just PAH_{3.3}: why galaxies turn red in the Near-Infrared, *Astronomy and Astrophysics*, 693, A204 (2025)
4. L. Napolitano,..., **S. Mascia**, et al., The dual nature of GHZ9: a remote X-ray source balancing AGN and star formation at $z=10.145$, submitted to *ApJ* (October 2024)
5. L. Napolitano,..., **S. Mascia**, et al., Seven wonders of Cosmic Dawn: JWST confirms a high abundance of galaxies and AGNs at $z \simeq 9 - 11$ in the GLASS field, *Astronomy and Astrophysics*, 693, A50 (2025)
6. E. Merlin,..., **S. Mascia**, et al., ASTRODEEP-JWST: NIRCам-HST multiband photometry and redshifts for half a million sources in six extragalactic deep fields, *Astronomy and Astrophysics*, 691, A240 (2024)
7. H. Weida,..., **S. Mascia**, et al., Characterizing the Average Interstellar Medium Conditions of Galaxies at $z \sim 5.6 - 9$ with Ultraviolet and Optical Nebular Lines, *The Astrophysical Journal*, 971, 21, (2024)

8. A. Calabrò,..., **S. Mascia**, et al., Evidence of extreme ionization conditions and low metallicity in GHZ2/GLASS-z12 from a combined analysis of NIRSpec and MIRI observations, *The Astrophysical Journal*, Volume 975, 2, 245 (2024)
9. M. Castellano,..., **S. Mascia**, et al., JWST NIRSpec Spectroscopy of the Remarkable Bright Galaxy GHZ2/GLASS-z12 at Redshift 12.34, *The Astrophysical Journal*, 972, 143 (2024)
10. A. Calabrò,..., **S. Mascia**, et al., The evolution of the SFR and Σ SFR of galaxies in cosmic morning ($4 < z < 10$), *Astronomy and Astrophysics*, 690, A290 (2024)
11. L. Napolitano,..., **S. Mascia**, et al., Peering into cosmic reionization: the Ly α visibility evolution from galaxies at $z = 4.5 - 8.5$ with JWST, *Astronomy and Astrophysics*, 688, A106 (2024)
12. J. W. Cole,..., **S. Mascia**, et al., CEERS: Increasing Scatter along the Star-Forming Main Sequence Indicates Early Galaxies Form in Bursts, *The Astrophysical Journal*, Volume 979, 2, 193
13. K. Boyett,..., **S. Mascia**, et al., A massive interacting galaxy 525 million years after the Big Bang, *Nat Astron* (2024)
14. A. Calabrò,..., **S. Mascia**, et al., Near-infrared emission line diagnostics for AGN from the local Universe to $z \sim 3$, *Astronomy and Astrophysics* 679, A80 (2023)
15. L. Napolitano,..., **S. Mascia**, et al., Identifying Ly α emitter candidates with Random Forest: learning from galaxies in CANDELS survey, *Astronomy and Astrophysics* 677, A138 (2023)
16. M. Castellano,..., **S. Mascia**, et al., The ionizing photon production efficiency of bright $z \sim 2-5$ galaxies, *Astronomy and Astrophysics*, 675, A121 (2023)
17. N. Roy,..., **S. Mascia**, et al., Early Results from GLASS-JWST XXII: Rest frame UV-optical spectral properties of Lyman-alpha emitting galaxies at $3 < z < 6$, *The Astrophysical Journal Letters*, 952, L14 (2023)
18. M. Talia,..., **S. Mascia**, et al., The VANDELS ESO public spectroscopic survey: the spectroscopic measurements catalogue, *Astronomy and Astrophysics*, 678, A25 (2023)
19. T. Jones,..., **S. Mascia**, et al., Early Results from GLASS-JWST. XXI: Rapid assembly of a galaxy at $z = 6.23$ revealed by its C/O abundance, *The Astrophysical Journal*, 951, L17 (2023)
20. D. Paris,..., **S. Mascia**, et al., The GLASS-JWST Early Release Science Program. II. Stage I release of NIRCам imaging and catalogs in the Abell 2744 region, *The Astrophysical Journal*, Volume 952, 20 (2023)
21. B. Vulcani,..., **S. Mascia**, et al., Early results from GLASS-JWST. Unveiling a population of “red-excess” galaxies in Abell2744 and in the coeval field, *The Astrophysical Journal*, 948, L15 (2023)
22. M. Castellano,..., **S. Mascia**, et al., A High Density of Bright Galaxies at $z \approx 10$ in the A2744 region, *The Astrophysical Journal*, 948, L14 (2023)
23. T. Morishita,..., **S. Mascia**, et al., Early results from GLASS-JWST. XVIII: A spectroscopically confirmed protocluster 650 million years after the Big Bang, *The Astrophysical Journal*, 947, L24 (2023)
24. G. Prieto-Lyon,..., **S. Mascia**, et al., The production of ionizing photons in UV-faint $z \sim 3-7$ galaxies, *Astronomy and Astrophysics*, 672, A186 (2023)

25. T. Treu,..., **S. Mascia**, et al., Early results from GLASS-JWST. XII: the morphology of galaxies at the epoch of reionization, *The Astrophysical Journal*, 942, L28 (2023)
26. P. Santini,..., **S. Mascia**, et al., Early results from GLASS-JWST. XI: Stellar masses and mass-to-light ratio of $z > 7$ galaxies, *The Astrophysical Journal*, 942, L27 (2023)
27. N. Leethochawalit,..., **S. Mascia**, et al., Early results from GLASS-JWST. X: Rest-frame UV-optical properties of galaxies at $7 < z < 9$, *The Astrophysical Journal*, 942, L26 (2023)
28. A. Calabrò,..., **S. Mascia**, et al., The properties of the Interstellar Medium in star-forming galaxies at redshifts $2 < z < 5$ from the VANDELS survey, *Astronomy and Astrophysics*, 667, A117 (2022)
29. M. Castellano,..., **S. Mascia**, et al., Early results from GLASS-JWST. III: Galaxy candidates at $z \sim 9$ -15, *The Astrophysical Journal*, 938, L15 (2022)
30. G. Roberts-Borsani,..., **S. Mascia**, et al., Early Results from GLASS-JWST. I: Confirmation of Lensed $z > 7$ Lyman-Break Galaxies Behind the Abell 2744 Cluster With NIRISS, *The Astrophysical Journal*, 938, L13 (2022)
31. A. Saxena,..., **S. Mascia**, et al., Strong C IV emission from star-forming galaxies: a case for high Lyman continuum photon escape, *Monthly Notices of the Royal Astronomical Society*, 517, 1098 (2022)
32. T. Treu,..., **S. Mascia**, et al., The GLASS James Webb Space Telescope Early Release Science Program. I. Survey Design and Release Plans, *The Astrophysical Journal*, 935, 110 (2022)